Asymmetric Bonding and Buyer Dominance: Two Composing Mechanisms for Self-Enforcing N-Party Coordination

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Abstract

We present FIGARO, a settlement mechanism that enables self-enforcing agreements between mutually distrusting parties without arbitrators, timeouts, or governance. The mechanism composes two distinct primitives. *Asymmetric bonding*: each party locks collateral equal to twice the transaction value, making cooperation the weakly dominant strategy in a one-shot game and the unique profile surviving iterated elimination of weakly dominated strategies. *Buyer dominance with atomic resolution*: only the buyer can trigger resolution, and resolution settles all orders in a process simultaneously or not at all. The two mechanisms are inseparable: bonding alone gives independently secured bilateral edges that cannot coordinate at the multi-party level; buyer dominance alone is without force absent the bonding equilibrium.

Asymmetric bonding scales naturally from two-party exchange to N-party process chains: because each seller's bond is tied to the cumulative value of all upstream work rather than to the local payment, the Nash equilibrium is preserved at every chain position and the seller's risk-to-reward ratio grows with chain depth. Buyer dominance with atomic resolution then operates on the already-scaled mesh: the all-or-nothing resolution rule induces a weakest-link subgame among sellers. We show that the resulting peer-enforcement properties obtain under strictly weaker assumptions than the joint-liability mechanisms classically analyzed in the microfinance literature.

Keywords: mechanism design, Nash equilibrium, collateral bonding, multi-party coordination, process chains, peer enforcement

1 Introduction

1.1 The Problem

Economic coordination between strangers requires trust infrastructure. When two parties who have never met wish to exchange value, they face the classic commitment problem: either party may defect after the other has performed. The historical solution is *institutional intermediation*—firms, courts, escrow services, platform operators—each of which introduces costs, delays, jurisdictional dependencies, and power asymmetries.

The problem intensifies in multi-party settings. A food delivery involves a cook, a kitchen operator, an ingredient sourcer, and a courier; a procurement process involves engineers, manufacturers, inspectors, and logistics providers. The standard architectural response is to aggregate participants into a single legal entity that internalizes the coordination problem and resolves it through hierarchical management.

Blockchain-based smart contracts have produced a rich literature on trustless exchange, but most protocols address financial operations—swaps, lending, derivatives—rather than general coordination. Those that target coordination (e.g., [Lesaege et al., 2019, Aragon, 2017]) typically reintroduce the intermediary as an on-chain dispute resolution mechanism: a jury, a governance body, or a timeout that unilaterally releases funds.

1.2 Our Contribution

We present FIGARO, which achieves self-enforcing coordination through two composing mechanisms: *asymmetric bonding*, which produces the bilateral Nash equilibrium and scales it naturally to N-party process chains by tying each seller's bond to cumulative upstream value; and *buyer dominance with atomic resolution*, which enforces inter-seller coordination on the already-scaled mesh. The two mechanisms are inseparable: bonding alone yields independently secured edges that cannot coordinate at the multi-party level; buyer dominance alone is worthless without the bonding equilibrium. Together they make the mesh resolvable from a single signature with cooperation pressure propagating through it, and they create the conditions for participants to resolve performance among themselves as social agents: the bonded commitment puts buyer and sellers in a position where working out difficulties between them is the dominant strategy, and external dispute resolution is a marginal residual rather than the primary path (Section 6). The contributions are:

(i) **A minimal settlement primitive** with no timeout, no governance, and no escape hatches from the bonded state. We show that this minimality is not a simplification but a *security requirement*: every additional exit path weakens the Nash equilibrium (Theorem 4.7).

(ii) **Asymmetric bonding scales to N-party chains.** We prove (Theorem 5.5) that cooperation remains weakly dominant at every position in an arbitrary-length process chain (with strict dominance whenever at least one other player's cooperation is not already foreclosed), and is the unique profile surviving iterated elimination of weakly dominated strategies; the seller's risk-to-reward ratio increases with chain depth.

(iii) **Atomic resolution** as a coordination forcing function. We show that all-or-nothing process settlement induces a weakest-link subgame among sellers, creating endogenous peer pressure without explicit communication or governance.

(iv) **A reduction over joint-liability microfinance.** We prove (Theorem 5.15) that the peer-enforcement outcome common to Ghatak [1999] and Stiglitz [1990] is obtained by the bonded commitment mechanism under strictly weaker assumptions: no repeated interaction, no local information, no exogenous punishment technology, and no joint-liability contracting.

1.3 Paper Organization

Section 2 surveys related work. Section 3 presents the mechanism formally. Section 4 proves the game-theoretic properties. Section 5 extends the model to N-party process chains, including a comparison with joint-liability microfinance (Section 5.5). Section 6 treats dispute resolution as a marginal external concern. Section 7 concludes.

1.4 Architectural Posture

The mechanism is ideologically agnostic; the graph is the politics. A consequence of locating the whole of the present paper's argument at the mechanism level is that the mechanism admits no commitments about *what* is being coordinated or *under what normative regime.* Currency, jurisdiction, identity type, arbitration forum, role structure, price-discovery mechanism, contribution metric—none of these are in the mechanism; all of them are expressed in the *graph* that parties compose on top of it. A Dutch-auction-and-private-operator graph expresses a market-liberal coordination pattern; a cooperative-bonding-and-contribution-weighted-resolution graph expresses a cooperative one; an interest-free risk-sharing graph expresses an Islamic-finance one. The mechanism enables each and prefers none. This is not evasion. It is the architectural consequence of placing the enforcement function at TCP/IP altitude: a layer below which nothing remains to factor out, and a layer above which the normative choices become legible as choices rather than as properties of the infrastructure.

Composability is external-first. A second consequence of the mechanism's narrow construction is that its useful extensions are predominantly *external* rather than internal. The mechanism does not include a dispute forum (Kleros, the Singapore International Arbitration Centre, or any court); it does not include a carbon-offset market (Klima DAO, Toucan, Moss); it does not include a prediction market, an insurance pool, a lending facility, a tax-reporting service, an identity provider, a storage layer, or a messaging fabric. Each of these exists as a working external protocol or institution, and an assembly composed on top of the mechanism can name any of them in its graph: a forum-clause names the dispute forum the parties agree to; an attestation clause names the offset operator the buyer wants to compose; a sub-order carries the buyer's offset purchase to the named operator within the same atomic resolution as the original commerce. The mechanism itself need not (and, by Theorem 4.7, must not) absorb the bookkeeping that such composition implies. The composability posture is the architectural counterpart of the ideological-agnosticism claim above: the mechanism makes external surfaces composable; the graph names which externals it composes with.

2 Related Work

Mutual-destruction escrows. The direct ancestor of FIGARO is the Safe Remote Purchase contract [Solidity Team, 2016], included in the Solidity documentation as a reference pattern: buyer and seller each lock 2× the item value, and only mutual agreement releases the funds. This two-party, single-order pattern is the seed from which FIGARO grew. BitHalo [Zimbeck, 2014] independently explored mutual-destruction deposits for peer-to-peer Bitcoin trade.

The central problem was generalizing SRP beyond two parties. SRP uses symmetric 2× deposits with mutual agreement to release—both parties must cooperate. This works for a single exchange but cannot compose: in a twelve-seller delivery workflow, there is no natural "mutual agreement" among all participants. *Asymmetric bonding* (Theorem 3.1) solves this. By scaling each seller's bond with cumulative upstream value rather than fixing it per trade, the mechanism creates a mesh of independently secured edges. Each edge carries its own Nash equilibrium (Theorem 4.3), and the mesh decomposes as finely as is operationally tolerable.

Buyer dominance then takes advantage of this mesh structure to make multi-party coordination resolvable from a single signature. Because a single party resolves the entire process atomically, coordination pressure propagates through the mesh without any management layer (Theorem 5.11).

State channels and payment networks. The Lightning Network [Poon and Dryja, 2016] and Raiden [Raiden, 2017] use time-locked deposits for off-chain payment. These protocols optimize for high-frequency bilateral payments and rely on timeouts for dispute resolution—precisely the escape hatch that FIGARO eliminates (Theorem 4.7). More critically, state channels do not address multi-party *coordination* (delivery, fulfillment, service composition); they address multi-hop *payment routing*.

On-chain dispute resolution. Kleros [Lesaege et al., 2019] provides decentralized arbitration via Schelling-point juror selection. Aragon Court [Aragon, 2017] offers similar functionality within DAO governance. Both reintroduce a *discretionary human decision* into the resolution path, which breaks the self-enforcing property (Theorem 4.7, Case 2).

Optimistic mechanisms. Optimistic rollups [Optimism, 2021] and optimistic oracles [UMA, 2020] use a challenge-response pattern: an assertion stands unless challenged within a timeout window. This requires liveness assumptions (the challenger must be online) and timeout calibration (too short enables fraud; too long delays finality). FIGARO has no timeout from the bonded state—capital remains locked indefinitely until the buyer resolves, making liveness irrelevant to security.

Mechanism design. The revelation principle [Myerson, 1979] guarantees that any achievable outcome can be implemented via a direct mechanism. FIGARO's insight is that the "direct mechanism" for multi-party coordination need not involve message passing, reporting, or arbitration at all: locked capital *is* the mechanism. The closest theoretical antecedent is the concept of *deposit games* [Asgaonkar and Krishnamachari, 2019], which analyze two-party security deposits. We extend deposit-game analysis to N-party process chains under asymmetric bonding.

Joint-liability microfinance. Group lending models—most prominently the Grameen Bank [Yunus, 1999] and its formalizations by Ghatak [1999] and Stiglitz [1990]—show that joint liability combined with a social substrate can produce peer selection and peer monitoring outcomes. We return to these models in Section 5.5: FIGARO reproduces the same peer-enforcement outcome under strictly weaker assumptions, with no appeal to repeated interaction or local information.

3 The Two Mechanisms: Formal Definition

3.1 Notation and Setup

Consider two parties, a *buyer* B and a *seller* S, who wish to exchange a payment $P > 0$ (denominated in a unit of account) for goods or services. Both parties hold sufficient balances.

Definition 3.1 (Asymmetric Bonding)**.** An *asymmetric bond* for a transaction with payment P and cumulative value $G \geq P$ is a pair of deposits:

$$C_b = 2P \quad \text{(buyer bond),} \tag{1}$$

$$C_s = 2G \quad \text{(seller bond).} \tag{2}$$

For a root transaction (no upstream value), $G = P$, so both bonds equal $2P$. For sub-orders in a process chain, $G > P$, creating an asymmetry where the seller's bond exceeds the buyer's.

Definition 3.2 (Commitment)**.** A *commitment* is a tuple $\langle B, S, \tau, P, h, \sigma, \delta \rangle$ where B is the buyer identity, S the seller identity, τ the unit of account, P the payment, h a content hash of the underlying agreement, σ a cryptographic salt, and δ a signature expiry deadline. Both parties sign the commitment.

Definition 3.3 (Process)**.** A *process* Π is a tuple $\langle \mathrm{id}, B, \tau, G, n, \mathcal{O} \rangle$ where id is a deterministic identifier derived from the root commitment, B is the root buyer (constant across all orders), τ is the denomination (constant), G is a *monotonic accumulator* tracking total committed value, n is the count of active orders, and $\mathcal{O}$ is the set of active order identifiers.

Remark 3.4 (Monotonicity Assumption). We assume the underlying settlement layer maintains G as a monotonic accumulator: every successful `commit` on Π increases G by exactly the new order's payment P, and no operation decreases G. This is an assumption about the settlement layer's bookkeeping discipline, not a property derived from the mechanism. The equilibrium analyses that follow apply to any settlement substrate that satisfies it; an implementation must enforce the assumption for the analyses to apply. The mechanism itself is substrate-independent: it can be realised on a cryptographically settled state machine, on a clearing house, on a trusted ledger, or on any other settlement layer that maintains the monotone-accumulator discipline.

Definition 3.5 (Buyer Dominance)**.** A process Π exhibits *buyer dominance* when (i) extension of Π via `commit` requires a dual-signed commitment authenticated by B's signature, and (ii) only B may trigger `resolveProcess`. No other identity—seller, third party, administrator, or governance body—has authority to extend or resolve Π.

The two conditions gate on different identity surfaces. Condition (i) gates on the identity of the *signer*, not on the agent that conveys the message: an authorized delegate carrying B's signature may convey a `commit` on B's behalf without breaking buyer dominance, since the substrate verifies the signature and admits the commitment if and only if it recovers to B. Condition (ii), in contrast, gates on the identity of the *caller* that issues `resolveProcess` directly. In deployed settings B may delegate the resolve action only through mechanisms that bind delegated authority to B's identity at the substrate layer (multi-signature wallets, account abstraction, or key-recovery schemes), not through signature-only conveyance.

Definition 3.6 (Atomic Resolution)**.** Resolution is *atomic*: when the buyer calls `resolveProcess`, every active order in the process settles simultaneously, or none does. There is no partial resolution and no per-order resolution path. Combined with buyer dominance, atomic resolution is the second mechanism: it operates on the mesh of independently bonded edges induced by asymmetric bonding and makes multi-party coordination resolvable from a single signature, while inducing the inter-seller weakest-link subgame analyzed in Section 5.3.

Remark 3.7 (Single-Currency Binding). Every order in a process shares a single denomination τ (Theorem 3.3). This is not a notational convenience: the $2\times$ Nash-stability result of Theorem 4.3 relies on same-unit comparability between the buyer's bond $C_b = 2P$ and the seller's bond $C_s = 2G$. Mixing denominations within one process would require an external rate of substitution (an oracle, an exchange, or a pre-agreed conversion rate), each of which introduces a discretionary actor outside the bond structure and re-opens the escape-hatch problem of Theorem 4.7. Multi-denomination workflows are accordingly handled outside the mechanism, by parties converting into the process denomination before issuing a `commit`.

3.2 Protocol Operations

The mechanism exposes exactly two state-changing operations.

Operation 1: `commit`. Given a commitment c signed by both B and S:

1. Verify both signatures.

2. If c.processId $= 0$, derive a new process identifier from the commitment digest and initialize the process with $G \leftarrow P$ and $n \leftarrow 1$.

3. Otherwise verify that c.processId identifies an existing process, that the buyer field $c.B$ matches the process root buyer $\Pi.B$ and that the buyer signature recovers to $c.B$ (*buyer dominance*), that the denomination $c.\tau$ matches $\Pi.\tau$ (*single-currency binding*, Theorem 3.7), and that c.expectedCumulativeValue $= \Pi.G + P$ (*monotonicity*, Theorem 3.4).

4. Pull $C_b = 2P$ from B and $C_s = 2c$.expectedCumulativeValue from S.

5. Create or extend the process accordingly, update the monotonic accumulator, and emit an evidence event that fully records the commitment.

Operation 2: `resolveProcess`**.** Given a process identifier and the complete list of active commitments:

1. Verify the caller identity equals $\Pi.B$ (buyer dominance).
2. Verify that the list length equals $\Pi.n$ (atomic resolution: *all* orders must be included).
3. For each order o_i with payment P_i and cumulative snapshot G_i:

$$\pi_{s,i} = 2G_i + P_i, \tag{3}$$
$$\pi_{b,i} = P_i. \tag{4}$$

4. Transfer $\pi_{s,i}$ to each seller, $\pi_{b,i}$ to the buyer.
5. Mark all orders as resolved.

Remark 3.8 (Conservation Law)*.* For each order, the total distributed equals total locked:

$$\pi_{s,i} + \pi_{b,i} = (2G_i + P_i) + P_i = 2G_i + 2P_i = C_{s,i} + C_{b,i}. \tag{5}$$

The escrow holds zero balance after resolution.

Remark 3.9 (No Escape Hatches)*.* There is no timeout, no cancel-after-commitment, no admin override, no governance vote, and no unilateral exit from the bonded state. Once both parties have committed, the only path to fund recovery is buyer-initiated resolution. This is not a temporary simplification; it is a *security requirement.* Theorem 4.7 proves that any additional exit path weakens the Nash equilibrium.

3.3 Net Payoffs

Definition 3.10 (Generalized Payout Functions)**.** For a bond multiplier $k \geq 1$, an order with payment $P > 0$ and cumulative value $G \geq P$, define the *settlement payout functions*:

$$\pi_s(k, G, P) = k \cdot G + P, \tag{6}$$
$$\pi_b(k, P) = (k - 1) \cdot P. \tag{7}$$

These satisfy the conservation law $\pi_s + \pi_b = k \cdot G + k \cdot P = C_s + C_b$ for $C_s = k \cdot G$ and $C_b = k \cdot P$. The *net payoff* is the payout minus the deposited bond:

$$u_s^{\text{net}}(k) = \pi_s - C_s = P, \tag{8}$$
$$u_b^{\text{net}}(k) = \pi_b - C_b = -P. \tag{9}$$

Remark 3.11. For the mechanism's $k = 2$: $\pi_s = 2G + P$, $\pi_b = P$, $C_s = 2G$, $C_b = 2P$. The net payoffs ($+P$ for the seller, $-P$ for the buyer) are independent of k—the multiplier affects only the capital at risk, not the final value transfer.

4 Game-Theoretic Analysis

4.1 Two-Party Nash Equilibrium

Definition 4.1 (The Bonding Game)**.** The *bonding game* $\Gamma = (\{B, S\}, \{A_B, A_S\}, \{u_b, u_s\})$ is a one-shot, simultaneous-move, complete-information game:

- **Players**: buyer B, seller S.
- **Action sets**: $A_B = A_S = \{\text{C}, \text{D}\}$.

- **Interpretation**: C for S means delivering the agreed goods/services; C for B means calling `resolveProcess`. D means failing to perform.
- **Payoff functions** (net of bond deposits), for a root order with payment $P > 0$ and $G = P$:

$$u_b(a_B, a_S) = \begin{cases} -P & \text{if } a_B = \mathrm{C} \text{ and } a_S = \mathrm{C}, \\ -2P & \text{otherwise.} \end{cases} \tag{10}$$

$$u_s(a_B, a_S) = \begin{cases} +P & \text{if } a_B = \mathrm{C} \text{ and } a_S = \mathrm{C}, \\ -2G & \text{otherwise.} \end{cases} \tag{11}$$

The "otherwise" payoff is identical across all non-cooperative outcomes because any defection prevents resolution: bonds remain locked indefinitely, yielding a net loss equal to the full deposit.

Table 1: Payoff matrix for the bonding game Γ. Entry format: (u_b, u_s). Parameters: $P > 0$, $G \geq P > 0$.

	$a_B = \mathrm{C}$	$a_B = \mathrm{D}$
$a_S = \mathrm{C}$	$(-P,\ +P)$	$(-2P,\ -2G)$
$a_S = \mathrm{D}$	$(-2P,\ -2G)$	$(-2P,\ -2G)$

Definition 4.2 (Dominance)**.** Action a_i *weakly dominates* a'_i for player i if $u_i(a_i, a_{-i}) \geq u_i(a'_i, a_{-i})$ for all a_{-i}, with strict inequality for at least one a_{-i}. If the inequality is strict for *all* a_{-i}, the dominance is *strict.*

Theorem 4.3 (Two-Party Nash Equilibrium)**.** *In the bonding game* Γ *of Theorem 4.1:*

(a) C *weakly dominates* D *for both the buyer and the seller.*

(b) (C, C) *is the unique profile surviving iterated elimination of weakly dominated strategies. The profile* (D, D) *also satisfies the Nash condition (no* profitable *deviation, since both yield the locked-bond payoff), but is weakly dominated and is removed in the first elimination round; we adopt the iterated-elimination-of-weakly- dominated-strategies (IEWDS) refinement, standard for mechanism design [Myerson, 1979], as the operative equilibrium concept.*

(c) (C, C) *is the unique Pareto-optimal outcome.*

Proof. **Part (a): Weak dominance.**

Buyer. We compare $u_b(\mathrm{C}, a_S)$ vs. $u_b(\mathrm{D}, a_S)$ for each seller action:

$$a_S = \mathrm{C}: \quad u_b(\mathrm{C}, \mathrm{C}) = -P > -2P = u_b(\mathrm{D}, \mathrm{C}), \tag{12}$$

$$a_S = \mathrm{D}: \quad u_b(\mathrm{C}, \mathrm{D}) = -2P = u_b(\mathrm{D}, \mathrm{D}). \tag{13}$$

Inequality (12) is strict (since $P > 0$); (13) is an equality. By Theorem 4.2, C weakly dominates D for B.

Seller. We compare $u_s(a_B, \mathrm{C})$ vs. $u_s(a_B, \mathrm{D})$ for each buyer action:

$$a_B = \mathrm{C}: \quad u_s(\mathrm{C}, \mathrm{C}) = +P > -2G = u_s(\mathrm{C}, \mathrm{D}), \tag{14}$$

$$a_B = \mathrm{D}: \quad u_s(\mathrm{D}, \mathrm{C}) = -2G = u_s(\mathrm{D}, \mathrm{D}). \tag{15}$$

Inequality (14) is strict (since $P > 0$ and $G > 0$ imply $P + 2G > 0$); (15) is an equality. C weakly dominates D for S.

Part (b): Unique Nash equilibrium. Since C weakly dominates D for both players, neither can profitably deviate from (C, C). We verify no other profile is a Nash equilibrium:

- (C, D): Seller deviates to C, gaining $P - (-2G) = P + 2G > 0$. Not a NE.
- (D, C): Buyer deviates to C, gaining $-P - (-2P) = P > 0$. Not a NE.
- (D, D): Buyer deviates to C with no change (both yield $-2P$). Seller deviates to C with no change (both yield $-2G$). This profile satisfies the Nash condition (no *profitable* deviation), but an iterated elimination of weakly dominated strategies removes D for both players, leaving (C, C) as the unique survivor.

Part (c): Pareto optimality. The payoff vector $(-P, +P)$ Pareto-dominates $(-2P, -2G)$ since $-P > -2P$ and $+P > -2G$. □

Remark 4.4 (On Weak vs. Strict Dominance). The dominance is weak, not strict, because when the counterparty defects, both actions yield the same locked-capital payoff. This is intrinsic to the mechanism: once one party defects, the outcome is fully determined regardless of the other's choice. The critical property is that (D, D) does not survive iterated elimination of weakly dominated strategies—the standard refinement for mechanism design [Myerson, 1979].

4.2 Sufficiency of the $2\times$ Multiplier

We now characterize the multiplier k in the generalized bonding game where $C_s = k{\cdot}G$, $C_b = k{\cdot}P$, and payouts are given by Theorem 3.10.

Lemma 4.5 (Buyer Indifference at $k = 1$)**.** *For $k = 1$, the buyer is indifferent between cooperating and defecting when the seller cooperates: $u_b(\mathrm{C}, \mathrm{C}) = u_b(\mathrm{D}, \mathrm{C}) = -P$.*

Proof. With $k = 1$: $C_b = P$ and $\pi_b = (k-1)P = 0$ (from (7)). Under our accounting convention (Theorem 3.10), the buyer's net payoff under cooperation is $u_b(\mathrm{C}, \mathrm{C}) = \pi_b - C_b = 0 - P = -P$. Under defection, the buyer's bond remains locked: $u_b(\mathrm{D}, \mathrm{C}) = -C_b = -P$. Since $u_b(\mathrm{C}, \mathrm{C}) = u_b(\mathrm{D}, \mathrm{C}) = -P$, the buyer has no strictly profitable deviation from D—the mechanism provides zero marginal incentive to resolve. □

Theorem 4.6 (Minimal Viable Bond Multiplier)**.** *Let $k \in \mathbb{R}_{\geq 1}$ be the bond multiplier. Then:*

(a) For $k = 1$, C *does not weakly dominate* D *for the buyer: the buyer is indifferent when the seller cooperates.*

(b) For every $k > 1$, C *weakly dominates* D *for both players (Theorem 4.3); $k = 1$ is therefore the infimum of the multipliers for which weak dominance fails on the buyer side.*

(c) Among integer multipliers, $k = 2$ is the smallest value at which weak dominance holds. For any $k > 2$, the equilibrium properties are identical to $k = 2$, but capital requirements increase by $(k - 2)(G + P)$ per order with no improvement in dominance relations.

Thus $k = 2$ is the minimal integer *multiplier that yields weak dominance of* C *for both players, and is Pareto-optimal in capital efficiency among all integer multipliers; the underlying real-line cutoff is $k = 1$. The integer restriction reflects the discrete denomination of bonded transfers in deployment, not a property of the equilibrium analysis.*

Proof. **Part (a).** See Theorem 4.5. With $k = 1$, the buyer recovers $\pi_b = 0$ upon resolution, having deposited $C_b = P$. The net monetary payoff from cooperating is $-P$; the net from defecting is also $-P$ (bond locked). The buyer cannot be motivated by the mechanism alone to resolve.

Part (b). For $k > 1$: $\pi_b = (k - 1)P$, $C_b = kP$. Cooperation yields $u_b(\mathrm{C}, \mathrm{C}) = (k - 1)P - kP = -P$; defection yields $u_b(\mathrm{D}, \mathrm{C}) = -kP$. The cooperation–defection gap is $-P - (-kP) = (k-1)P$, which is strictly positive for every $k > 1$. Weak dominance for the buyer therefore holds for every $k > 1$ and fails at $k = 1$ (the gap collapses to zero), establishing $k = 1$ as the infimum at which weak dominance fails on the buyer side. The seller-side weak dominance follows from Theorem 4.3(a) for any $k \geq 1$ (the seller's net payoff under cooperation is $+P$ regardless of k, while defection forfeits the bond kG).

Part (c). For integer k, the smallest value at which both Part (a) (failure at $k = 1$) and Part (b) (success for $k > 1$) admit a feasible bond is $k = 2$. The surplus capital locked at $k > 2$, $(k-2)P$ for the buyer and $(k-2)G$ for the seller, earns zero return and is pure deadweight cost. □

4.3 Escape-Hatch Weakness

Theorem 4.7 (Escape-Hatch Weakness). *Let Γ_E be the bonding game augmented with a unilateral exit action E available to player i, returning fraction $\alpha \in (0, 1]$ of player i's bond. Then either:*

(a) C *no longer weakly dominates* D *for at least one player (the Nash equilibrium of Theorem 4.3 may not survive), or*

(b) E requires a decision by a third party $J \notin \{B, S\}$ whose incentives are not constrained by the bond structure.

Proof. We analyze three exhaustive sub-cases of unilateral exit.

Case 1: Timeout (automatic exit after duration T). The action space becomes $A_B = \{\mathrm{C}, \mathrm{D}, \mathrm{Wait}\}$ where Wait means the buyer does not resolve and collects $\alpha \cdot C_b$ after time T. The buyer's payoff under Wait against a cooperating seller is:

$$u_b(\mathrm{Wait}, \mathrm{C}) = -C_b + \alpha \cdot C_b = -2P(1-\alpha). \tag{16}$$

Compare with cooperation: $u_b(\mathrm{C}, \mathrm{C}) = -P$. The buyer prefers C to Wait iff:

$$-P > -2P(1-\alpha) \iff 2(1-\alpha) > 1 \iff \alpha < \tfrac{1}{2}. \tag{17}$$

For $\alpha \geq \frac{1}{2}$, Wait yields $u_b \geq -P = u_b(\mathrm{C}, \mathrm{C})$, so C no longer (weakly) dominates Wait. The buyer may rationally wait out the timeout instead of resolving. Even for $\alpha < \frac{1}{2}$, the defection cost drops from $2P$ to $2P(1-\alpha)$, weakening the deterrent. The seller, anticipating the timeout, faces a modified game where the buyer's commitment to resolve is no longer credible—the timeout provides an alternative path to partial capital recovery, undermining the "no escape" property that sustains the equilibrium.

Case 2: Arbitrator override. A third party J can invoke E on behalf of either player. Formally, this extends the game to three players $\{B, S, J\}$. The mechanism's self-enforcing property requires that the payoff structure *alone* determines the outcome. Introducing J makes the outcome contingent on J's action, and J is not bonded: J deposits nothing, risks nothing, and has no mechanism-constrained incentive to decide correctly. Whether J is an individual (arbitrator), an algorithm (oracle), or a collective (jury), the resolution path now depends on an agent outside the bond structure. This is condition (b).

Case 3: Governance vote. A special case of Case 2 where J is replaced by a set of voters $\{J_1, \ldots, J_m\}$ who collectively decide on E. This inherits all problems of Case 2, plus: (i) coordination failures among voters, (ii) plutocratic capture when voting power is stake-weighted, (iii) voter apathy (rational ignorance). Each voter J_k is unbonded.

In all three cases, either C ceases to weakly dominate all alternatives (Case 1), or resolution depends on an unbonded external agent (Cases 2–3). □

Remark 4.8 (On Judicial Review and the Distinction from Cases 2–3). The theorem's "unbonded external agent" condition targets actors whose decision is *constitutive* of the resolution path internal to the mechanism—arbitrators, juries, oracles, voters embedded in the bonding game itself. It does not target the external legal forum that may, on the evidence record, adjudicate a dispute *outside* the bonding game under doctrines of duress, frustration, impossibility, unconscionability, or public policy. Those forums are constrained by their own institutional bond structures (appellate review, professional sanction, doctrinal consistency) and operate on the bonded commitment as evidentiary input rather than as a resolution path within the mechanism.

Corollary 4.9. *No timeout duration T and no recovery fraction $\alpha > 0$ can be added to the bonding game without weakening the buyer's incentive to cooperate. In particular, full recovery ($\alpha = 1$) reduces the game to a zero-cost option, making defection strictly dominant for the buyer.*

Remark 4.10 (Robustness to Weaker Rationality). The dominance results stated in monetary terms extend to a wider class of agent preferences than risk-neutral expected-payoff maximisation. Three observations. First, the dominance argument compares the payoff under cooperation ($+P_i$ for S_i, $-P_i$ for B) to the payoff under defection ($-2G_i$ for S_i, $-2P_i$ for B); since $P_i > -2G_i$ and $-P_i > -2P_i$ are preserved under any strictly monotone utility transformation, weak dominance of C holds for any preference order that prefers more money to less, including arbitrary risk-averse and loss-averse utility specifications. Second, the cooperation–defection gap $P_i + 2G_i$ is bounded away from zero (and grows with chain depth), so the dominance margin is not perturbation-sensitive: a trembling-hand refinement preserves $(\mathrm{C}, \ldots, \mathrm{C})$ as the unique surviving profile, since small perturbations to other players' actions cannot flip a strictly positive gap. Third, the dominance argument requires only one comparison per player and does not depend on iterated reasoning about counterparty rationality, common knowledge of payoffs, or beliefs about types. What remains genuinely outside the analysis is behaviour under non-pecuniary preferences—spite, fairness norms, or intrinsic motivation to defect—where the monetary payoff matrix does not capture all relevant utilities.

5 N-Party Process Chains

5.1 N-Party Asymmetric Bonding

We now extend the two-party model to an arbitrary-length chain of sellers.

Definition 5.1 (Process Chain)**.** A *process chain* is a sequence of bonded orders $\langle o_1, o_2, \ldots, o_n \rangle$ sharing a common buyer B and denomination τ. Order o_i has local payment $P_i > 0$ and cumulative value:

$$G_i = \sum_{j=1}^{i} P_j. \tag{18}$$

The seller of o_i posts bond $C_{s,i} = 2G_i$; the buyer posts bond $C_{b,i} = 2P_i$.

Definition 5.2 (N-Party Bonding Game)**.** The *N-party bonding game* $\Gamma_N = (\{B, S_1, \ldots, S_n\}, \{A_i\}, \{u_i\})$ extends Γ to n sellers. Each seller S_i has action set $\{\mathrm{C}, \mathrm{D}\}$. The buyer has action set $\{\mathrm{C}, \mathrm{D}\}$ and chooses after observing each seller's realized action; we assume perfect monitoring, so B perfectly observes whether each S_i delivered, and plays the conditional strategy "resolve iff all sellers cooperated." The perfect-monitoring assumption corresponds to the deployed setting, in which the buyer receives either the agreed goods or not, and the buyer's observation of delivery is prior to the buyer's `resolveProcess` call. Imperfect monitoring is addressed at composable surfaces above the mechanism (attestation, schema-typed witness records, third-party signal authorization), which lie outside the present paper's scope. Payoffs under perfect monitoring:

$$u_{S_i}(a_B, a_{S_1}, \ldots, a_{S_n}) = \begin{cases} +P_i & \text{if } a_B = \mathrm{C} \text{ and } a_{S_j} = \mathrm{C} \ \forall j, \\ -2G_i & \text{otherwise.} \end{cases} \tag{19}$$

The buyer's payoff is:

$$u_b(a_B, a_{S_1}, \ldots, a_{S_n}) = \begin{cases} -\sum_{i=1}^{n} P_i & \text{if } a_B = \text{C} \text{ and } a_{S_j} = \text{C } \forall j, \\ -\sum_{i=1}^{n} 2P_i & \text{otherwise.} \end{cases} \tag{20}$$

Remark 5.3 (Sequential vs Simultaneous Equivalence). The game Γ_N is presented in simultaneous-move form, but in deployment the events of `commit`, delivery, and `resolveProcess` are sequential. The two representations are equivalent under asymmetric bonding: each seller S_i's payoff function depends on S_i's own action and the buyer's eventual `resolve`/`not-resolve` decision, and not on the order in which other sellers acted or on the temporal position of S_i's `commit` relative to others'. Defection by any participant forces non-resolution (Theorem 3.6), which yields the same locked-bond payoff $-2G_i$ to S_i regardless of when in the sequence the defection occurs. The dominance argument of Theorem 4.3 therefore applies edge-by-edge at every node of the corresponding extensive-form game, so the IEWDS profile of the simultaneous-move representation coincides with the subgame-perfect Nash equilibrium of the extensive-form game. We use the simultaneous-move representation for expositional clarity.

Lemma 5.4 (Buyer Best Response). *The buyer's* post-commit *action set in Γ_N is $A_B = \{resolve, not\text{-}resolve\}$ (the buyer's prior* `commit` *actions establish the chain and are fixed at the point this lemma applies). Under perfect monitoring, the buyer's best response to the cooperative seller profile* $(\text{C}, \ldots, \text{C})$ *is resolve. Under any profile in which at least one seller defects, the buyer is indifferent between resolve and not-resolve (both yield the locked-bond payoff $-\sum_i 2P_i$).*

Proof. The buyer's only post-commit decision is whether to call `resolveProcess`; there is no continuation action that strictly improves on $-\sum_i 2P_i$ once any seller has defected, by atomic resolution (Theorem 3.6). Under $(\text{C}, \ldots, \text{C})$, the buyer's payoff under resolve is $-\sum_i P_i$ versus $-\sum_i 2P_i$ under not-resolve; the cooperation–defection gap $\sum_i P_i > 0$ is the linear sum of the bilateral gaps of Theorem 4.3(a), applied to each order independently. Hence resolve strictly dominates not-resolve on the cooperative profile, and the two are payoff-equal off it. □

Theorem 5.5 (N-Party Nash Equilibrium). *In Γ_N, cooperation is a weakly dominant strategy for every seller S_i at every position $i \in \{1, \ldots, n\}$, and* $(\text{C}, \text{C}, \ldots, \text{C})$ *is the unique outcome surviving iterated elimination of weakly dominated strategies.*

Proof. Fix an arbitrary seller S_i. Consider the two cases for the remaining players' joint action profile a_{-i}.

Case 1: All other sellers cooperate and buyer resolves. Then:

$$u_{S_i}(\ldots, \text{C}_i, \ldots) = +P_i, \tag{21}$$
$$u_{S_i}(\ldots, \text{D}_i, \ldots) = -2G_i. \tag{22}$$

Since $P_i > 0$ and $G_i > 0$, we have $P_i > -2G_i$, i.e., $P_i + 2G_i > 0$. The cooperation payoff strictly exceeds the defection payoff.

Case 2: At least one other seller defects, or buyer defects. Then resolution does not occur regardless of S_i's action:

$$u_{S_i}(\ldots, \text{C}_i, \ldots) = u_{S_i}(\ldots, \text{D}_i, \ldots) = -2G_i. \tag{23}$$

Payoffs are equal.

Combining both cases, C weakly dominates D for S_i, with strict inequality in Case 1. Since the choice of i was arbitrary, this holds for all n sellers. The buyer's dominance follows from Theorem 5.4: on the cooperative seller profile the buyer's best response is resolve, with payoff $-\sum P_i$ strictly exceeding the payoff $-2\sum P_i$ under not-resolve.

Iterated elimination: in the first round, D is eliminated for all players (weakly dominated). The unique surviving profile is $(\text{C}, \ldots, \text{C})$. □

Remark 5.6 (Strength of the N-Party Result). The seller's dominance condition, $P_i + 2G_i > 0$, holds with a margin that *grows* with chain depth (since $G_i = \sum_{j \le i} P_j$ is monotonic). The cooperation-defection payoff gap for S_i is:

$$\Delta_i = P_i - (-2G_i) = P_i + 2G_i = P_i + 2\sum_{j=1}^{i} P_j \ge 3P_i. \tag{24}$$

The gap is at least $3P_i$ (equality only at the root where $G_i = P_i$) and grows with every upstream payment. This is the formal basis of the "geometric coordination pressure" in Theorem 5.7.

Corollary 5.7 (Coordination Pressure Grows With Depth)**.** *The risk-to-reward ratio for seller S_i is:*

$$\rho_i = \frac{C_{s,i}}{P_i} = \frac{2G_i}{P_i} = \frac{2\sum_{j=1}^{i} P_j}{P_i}. \tag{25}$$

For equal payments ($P_j = p$ for all j), $\rho_i = 2i$, growing linearly with chain depth. For value chains where early stages contribute disproportionately ($P_1 \gg P_n$, with P_n bounded below), ρ_n scales as $2G_n/P_n$ and grows faster than linearly in n relative to the late-stage contribution. The qualitative point is that the deeper participant absorbs the cumulative upstream risk; the precise growth rate depends on the shape of the payment schedule.

Example 5.8. Alice orders food (\$10) from Bob, who needs delivery (\$2) from Charlie:

Party	Role	P_i	G_i	$C_{s,i}$	ρ_i
Bob	Seller 1	\$10	\$10	\$20	2.0
Charlie	Seller 2	\$2	\$12	\$24	12.0

Charlie risks \$24 to earn \$2 ($\rho = 12$). Bob risks \$20 to earn \$10 ($\rho = 2$). The deeper participant faces amplified consequences.

5.2 Self-Enforcing Cumulative-Value Reporting

A natural question: must the mechanism validate the cumulative value reported by each seller at commit time? It need not. The bonding rule itself makes honest reporting the dominant strategy whenever the probability of non-resolution is positive.

Theorem 5.9 (Cumulative-Value Reporting Honesty)**.** *In any game where seller S_i chooses a reported cumulative value $G'_i \ge G_i$ (with G_i being the true value maintained by the settlement layer's accumulator under Theorem 3.4), truth-telling ($G'_i = G_i$) weakly dominates all alternatives. The dominance is strict whenever the probability of non-resolution is positive.*

Proof. By Theorem 3.4 the accumulator constraint $G'_i \ge G_i := \Pi.G_{\text{prev}} + P_i$ holds: any report with $G'_i < G_i$ is rejected by the settlement layer at `commit` time and never enters the accumulator (cf. Theorems 3.2 and 3.3, the `expectedCumulativeValue` clause). We therefore restrict attention to $G'_i \ge G_i$.

Let $q \in [0, 1]$ be the probability that the buyer resolves the process (a function of all sellers' performance, exogenous to S_i's reporting decision). Define S_i's expected payoff as a function of the report G'_i:

$$\mathbb{E}[u_{S_i}(G'_i)] = q \cdot (\underbrace{2G'_i + P_i}_{\text{payout}} - \underbrace{2G'_i}_{\text{bond}}) + (1-q) \cdot (-2G'_i) = q \cdot P_i - (1-q) \cdot 2G'_i. \tag{26}$$

The derivative with respect to G'_i is:

$$\frac{\partial}{\partial G'_i} \mathbb{E}[u_{S_i}] = -2(1-q). \tag{27}$$

For $q < 1$ (non-zero probability of non-resolution), this derivative is strictly negative: expected utility is strictly decreasing in the reported cumulative value. The optimum is therefore at the smallest feasible value, $G_i' = G_i$.

For $q = 1$ (certain resolution), the derivative is zero and the seller is indifferent—overstating wastes capital but does not affect the net payoff. Truth-telling remains weakly optimal.

Since $q < 1$ in any realistic setting (the seller cannot guarantee other sellers' cooperation), truth-telling strictly dominates overstating. □

Remark 5.10 (Why the Mechanism Does Not Validate Cumulative-Value Reports). The theorem licenses a deliberate omission in the mechanism: beyond enforcing monotonicity of the accumulator, the mechanism performs no validation of the cumulative-value reports it receives. Misreporting cannot profit a seller because the bonding rule already makes overstating G_i strictly dominated.

5.3 Atomic Resolution and Seller Coordination

Atomic resolution was defined formally in Theorem 3.6; we now develop its game-theoretic consequence.

Theorem 5.11 (Endogenous Coordination Pressure)**.** *The N-party bonding game under atomic resolution induces a* weakest-link public goods game *among sellers, where each seller's payout depends on universal cooperation.*

Proof. Consider the sellers' subgame (fixing the buyer's strategy at: resolve iff all sellers cooperate). Each S_i has payoff u_{S_i} as in (19).

Define the *externality* imposed by S_k's defection on S_i ($i \neq k$):

$$\varepsilon_{k \to i} = u_{S_i}(\text{all C}) - u_{S_i}(S_k \text{ defects}) = P_i - (-2G_i) = P_i + 2G_i. \tag{28}$$

This is strictly positive for all i, k with $i \neq k$. Every defection imposes a loss on every other seller. The total externality of S_k's defection is:

$$\mathcal{E}_k = \sum_{i \neq k} \varepsilon_{k \to i} = \sum_{i \neq k} (P_i + 2G_i). \tag{29}$$

This structure satisfies the defining properties of a *weakest-link game* [Hirshleifer, 1983]:

1. The group outcome depends on the minimum individual effort (one defection collapses all payoffs to the non-cooperative level).

2. The cooperative outcome $(\mathrm{C}, \ldots, \mathrm{C})$ is Pareto optimal among the subgame's Nash equilibria; the all-defect profile satisfies the Nash condition (no profitable deviation from a locked-bond payoff) but is removed by IEWDS, leaving $(\mathrm{C}, \ldots, \mathrm{C})$ as the unique IEWDS-surviving profile.

3. The action structure is *weak-link best-response coupling under perfect monitoring*: S_i's cooperation–defection gap is $P_i + 2G_i$ when every other seller and the buyer cooperate, and zero otherwise. The marginal value of cooperation is therefore not monotonically increasing in others' cooperation (the standard strategic-complementarity shape); it is constant on the all-cooperate profile and zero elsewhere, which is the characteristic structure of weak-link coordination rather than strategic complementarity.

Under one-shot, anonymous interaction, this externality structure suffices for the cooperation-pressure result of Theorem 5.15: every S_i has a direct mechanism-level economic interest of magnitude $P_i + 2G_i$ in S_k's cooperation, computable from the mechanism's monotonic accumulator alone. No information acquisition, repeated interaction, or social-substrate communication is required for the externality to be present; whether parties additionally engage in such practices in deployment is orthogonal to the equilibrium argument. □

Remark 5.12 (Distinction from the Experimental Weakest-Link Literature). The experimental weakest-link literature, originating with Van Huyck et al. [1990], finds that weakest-link games empirically produce coordination *failure* in groups larger than two: players converge to the Pareto-inferior all-defect equilibrium via pessimistic beliefs about co-players. This might appear to threaten the result above. It does not. The classical experimental result arises when cooperation is *collectively* preferred but not *individually* dominant — a setting where pessimistic beliefs about others are sufficient to rationalise defection. In the bonded commitment mechanism, defection is individually dominated regardless of what others do (Case 2 of the proof of Theorem 5.5: even when all others defect, S_i incurs $-2G_i$ whether it cooperates or defects, so there is no strategic incentive to defect). The cooperation pressure identified in Theorem 5.11 therefore arises in a setting where coordination failure through pessimistic beliefs cannot occur: each seller's dominant strategy is already to cooperate, and the weakest-link structure adds the externality characterisation on top of that result, not as a substitute for it.

Proposition 5.13 (Collusion is Strictly Unprofitable)**.** *Let $K \subseteq \{S_1, \dots, S_n\}$ be any non-empty coalition of sellers, and let $X \notin \{B, S_1, \dots, S_n\}$ be any external party who pays a side transfer $T \geq 0$ to K in exchange for joint defection by every member of K. Joint defection by K is not a Pareto improvement over universal cooperation for the coalition $K \cup \{X\}$: the minimum side transfer at which K's members are individually willing to defect strictly exceeds the maximum monetary benefit X can obtain from B's non-resolution. The dominance margin is at least $3\sum_{i\in K} P_i$ versus a benefit upper-bounded by $3\sum_{i\in K} P_i$ at the root order, with the gap strictly positive whenever K contains any non-root seller.*

Proof. For each $S_i \in K$, the cooperation–defection gap is $\Delta_i = P_i - (-2G_i) = P_i + 2G_i$ (Theorem 5.5, Case 1). For S_i to weakly accept the side payment, X must transfer at least Δ_i to S_i. The minimum total side transfer is therefore

$$T_{\min}(K) = \sum_{i\in K} \Delta_i = \sum_{i\in K} (P_i + 2G_i) \geq 3 \sum_{i\in K} P_i, \tag{30}$$

with equality only when every $S_i \in K$ is a root order ($G_i = P_i$).

X's monetary benefit from K's joint defection is bounded by the most an arbitrary external party with spite preferences could rationally pay for the buyer's harm. That harm has two components. Non-resolution locks B's bonds, totalling $\sum_{i=1}^{n} 2P_i$, and denies B the goods themselves; the value of the goods to B is bounded above by B's willingness to pay, $\sum_{i=1}^{n} P_i$ (the participation constraint that says B entered the bonding game). Restricting to the colluded orders, X's upper bound on benefit is

$$B_{\max}(K) \leq \sum_{i\in K} 2P_i + \sum_{i\in K} P_i = 3 \sum_{i\in K} P_i. \tag{31}$$

Combining the bounds, $T_{\min}(K) \geq 3\sum_{i\in K} P_i \geq B_{\max}(K)$, with strict inequality on the first step whenever K contains a seller S_i with $G_i > P_i$ (i.e., any non-root seller). For collusion to be Pareto-improving for $K \cup \{X\}$, X would need benefit strictly greater than the side-payment cost; this is impossible. Joint defection is therefore strictly Pareto-dominated. □

Remark 5.14 (Margin Grows with Chain Depth). The dominance margin $T_{\min}(K) - B_{\max}(K) \geq 2\sum_{i\in K}(G_i - P_i) \geq 0$ is monotone in chain depth: the deeper the colluded sellers sit in the chain, the larger the spread between their bonds $2G_i$ and the harm imposable on the buyer. Coalitions of deep-chain sellers are accordingly the most strictly unprofitable.

5.4 Seller Non-Performance and Externality Handling

A natural question in any multi-party coordination mechanism is how externalities are handled. Four categories arise, and the bonded commitment mechanism treats them with a single primitive — bonded sub-orders, optionally layered with external risk products — without requiring additional governance or protocol modifications.

(i) *Intra-process externalities*—the loss one seller's defection imposes on co-sellers—are handled by the bonding mechanism itself: defection is weakly dominated for every seller at every position (Theorem 5.5), and the cooperation pressure that propagates through atomic resolution (Theorem 5.11) is the externality made strategic. No additional handling is required.

(ii) *Seller non-performance*—the focus of the two paragraphs that follow—is absorbed either by the buyer committing a substitute order before resolution, or by an external risk product (performance guarantee, surety bond, insurance) layered over the bonded position.

(iii) *Buyer non-performance*—the buyer commits but never resolves—is absorbed by the no-escape-hatch property: the buyer's bond remains locked indefinitely, imposing a $2P$ deterrent equal to the buyer's worst-case loss (Theorem 3.9, Theorem 4.7). The sellers' bonds are likewise locked indefinitely; recovery for sellers in this case requires the same external risk products as in (ii).

(iv) *Third-party externalities*—harms imposed by the bonded process on non-participants (a courier causing a traffic accident, a kitchen polluting, a process generating a data leak)—compose into the mechanism the same way as (ii). The harmed party becomes a remediation buyer in a new bonded sub-order against the harming party as remediation seller; or an external risk product (liability insurance, pollution bond) layers over the bonded process in advance. The mechanism is uniform across externality types: the same composition pattern handles internal and third-party externalities, with the harmed party simply being internal or external to the original process.

The two paragraphs below develop case (ii) in detail; cases (i), (iii), and (iv) are addressed by cross-reference and by the same composition pattern.

Seller substitution. If seller S_i fails to perform, the buyer holds a non-resolved process in which S_i's bond is locked but delivery has not occurred. Before calling `resolveProcess`, the buyer may commit a replacement order with a new seller S_i' at the same or renegotiated payment. S_i' bonds against the current cumulative value, enters the process under the standard bonding rule, and delivers. When the buyer resolves, S_i' receives full settlement. S_i, having failed to perform, forfeits their bond under the bonding equilibrium. The externality of S_i's non-performance is therefore absorbed by S_i's own capital loss, not distributed to co-sellers: other sellers' payouts are unaffected.

External risk products. Where seller substitution is impractical—latency-sensitive processes, highly specialised roles, or cases where a replacement cannot be sourced—standard risk instruments such as performance guarantees, surety bonds, and insurance can layer over the bonded process without modifying the mechanism. These instruments hold independently bonded positions on the seller's behalf; they do not call `commit` or `resolveProcess`, hold no mechanism state, and leave the bonding equilibrium intact. The mechanism is therefore not the only layer of externality protection available to parties; it is the settlement primitive over which such risk products compose.

5.5 Comparison with Joint-Liability Microfinance

The coordination pressure of Theorem 5.11 is operationally analogous to the peer-enforcement equilibrium documented in joint-liability microfinance, formalized by Ghatak [1999] and Stiglitz

[1990] on the Grameen Bank model [Yunus, 1999]. Those models obtain peer enforcement through a combination of joint-liability contracting and a social substrate (repeated interaction, local information, exogenous punishment via social sanction); the contract alone is insufficient, and the substrate does load-bearing enforcement work. The bonded commitment mechanism reproduces the same equilibrium-selection effect under strictly weaker assumptions.

Proposition 5.15 (Assumption Reduction on Cooperation Pressure)**.** *The* cooperation-pressure component *of the equilibrium common to Ghatak [1999] and Stiglitz [1990]—cooperative equilibrium selection driven by co-participant externalities—is obtained by the bonded commitment mechanism under strictly weaker assumptions. Specifically,* FIGARO *requires none of the following: (i) repeated interaction, (ii) local information among sellers, (iii) a punishment technology exogenous to the contract, nor (iv) joint-liability contracting.*

Proof. Theorem 5.5 establishes cooperation as the unique profile surviving iterated elimination of weakly dominated strategies in a one-shot game. The argument uses only the mechanism's payoff function and makes no appeal to continuation value, trigger strategies, or folk-theorem constructions; (i) is not required.

The externality $\varepsilon_{k\to i} = P_i + 2G_i$ (Theorem 5.11) is computed from quantities enforced by the settlement layer's monotonic accumulator, independent of any seller's information about co-sellers' types, actions, or identities; (ii) is not required.

The punishment for a co-seller's defection is the loss of S_i's own bond $C_{s,i} = 2G_i$, imposed directly by the mechanism through non-resolution. No external punishment technology is invoked; (iii) is not required.

Finally, each $C_{s,i}$ is posted individually by S_i against S_i's own cumulative-value snapshot, not against the group's aggregate obligation. Coupling across sellers arises from the buyer's atomic-resolution constraint (Theorem 3.6), not from the bond structure; (iv) is not required. □

The reduction is scoped to cooperation pressure. Two further microfinance results—peer selection on private types [Ghatak, 1999] and peer monitoring under principal information asymmetry [Stiglitz, 1990]—are not directly reproduced; both require additional structure above the bonded primitive (attested credentials, signal-contingent resolution rules) and lie outside the present paper's scope.

The reduction reframes rather than corrects the microfinance theorems: they remain sound under their assumptions, but the assumptions characterize one particular settlement environment in which the mechanism was historically deployed (reputation within a social network), not necessary conditions for endogenous peer monitoring. Once the settlement layer admits bilaterally signed, individually collateralized commitments with atomic resolution, the same outcome obtains among mutual strangers in a single shot with no social substrate.

6 Dispute Resolution

Dispute resolution is marginal in this mechanism by design. Defection is weakly dominated for every party at every position (Theorem 4.3, Theorem 5.5); the cooperation pressure that propagates through atomic resolution (Theorem 5.11) sharpens this further by making each seller's payout contingent on universal cooperation among co-sellers; and the reduction over joint-liability microfinance (Theorem 5.15) places the bonded mechanism in the same equilibrium-selection regime as group lending under strictly weaker assumptions. The bonded process itself absorbs almost all of the work that traditional arrangements externalise to courts, arbitrators, and platform adjudication.

The microfinance literature provides a qualitative empirical reference. Joint-liability group lending, beginning with the Grameen Bank, has produced low documented default rates under conditions (repeated interaction, local information, social sanction) that the bonded mechanism replaces with one-shot collateral and atomic resolution [Yunus, 1999, Ghatak, 1999]. The fraction of bonded processes that require external dispute resolution is therefore expected to be small, but the present mechanism does not predict the rate quantitatively: the microfinance default rate emerges from the combined effect of peer selection, peer monitoring, and cooperation pressure, only the last of which the bonded mechanism reproduces (Theorem 5.15). Calibration is an empirical matter for deployment data, not a theorem of the present paper.

For the marginal cases where internal pressure does not resolve a dispute, the bonded commitment is compatible with external legal and arbitral forums. Theorem 4.8 establishes the formal distinction: such forums adjudicate *outside* the bonding game on the evidence record produced by the mechanism, rather than serving as a constitutive resolution path within the mechanism. Parties may name a forum-clause in their assembly graph (Section 1.4) selecting Kleros, an arbitral body, a court, or any other forum [Lesaege et al., 2019]; the mechanism does not bind a forum, and the choice does not affect the bonding equilibrium. The composition is asymmetric in an important way: FIGARO composes with external forums as evidentiary consumers of the bonded record, but external forums cannot compose with FIGARO as a constitutive resolution path within the mechanism without breaking the equilibrium (Theorem 4.7, Case 2). The asymmetry is the formal expression of the design stance: dispute resolution is a marginal external concern, not an internal mechanism component.

7 Conclusion

We have presented FIGARO, a settlement primitive that achieves self-enforcing multi-party coordination through two composing mechanisms: *asymmetric bonding*, which produces the bilateral Nash equilibrium and scales it naturally to N-party process chains by tying each seller's bond to cumulative upstream value; and *buyer dominance with atomic resolution*, which enforces inter-seller coordination on the already-scaled mesh. The mechanism is minimal—no timeout, no governance, no escape hatches—and the game-theoretic properties are strong: under perfect monitoring (Theorem 5.2) and complete information, cooperation is weakly dominant at every position in an N-party process chain, and the cooperative profile is the unique outcome surviving iterated elimination of weakly dominated strategies. The mechanism reproduces the cooperation-pressure component of the peer-enforcement equilibrium in joint-liability microfinance under strictly weaker assumptions, with no appeal to repeated interaction, local information, or social sanction.

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